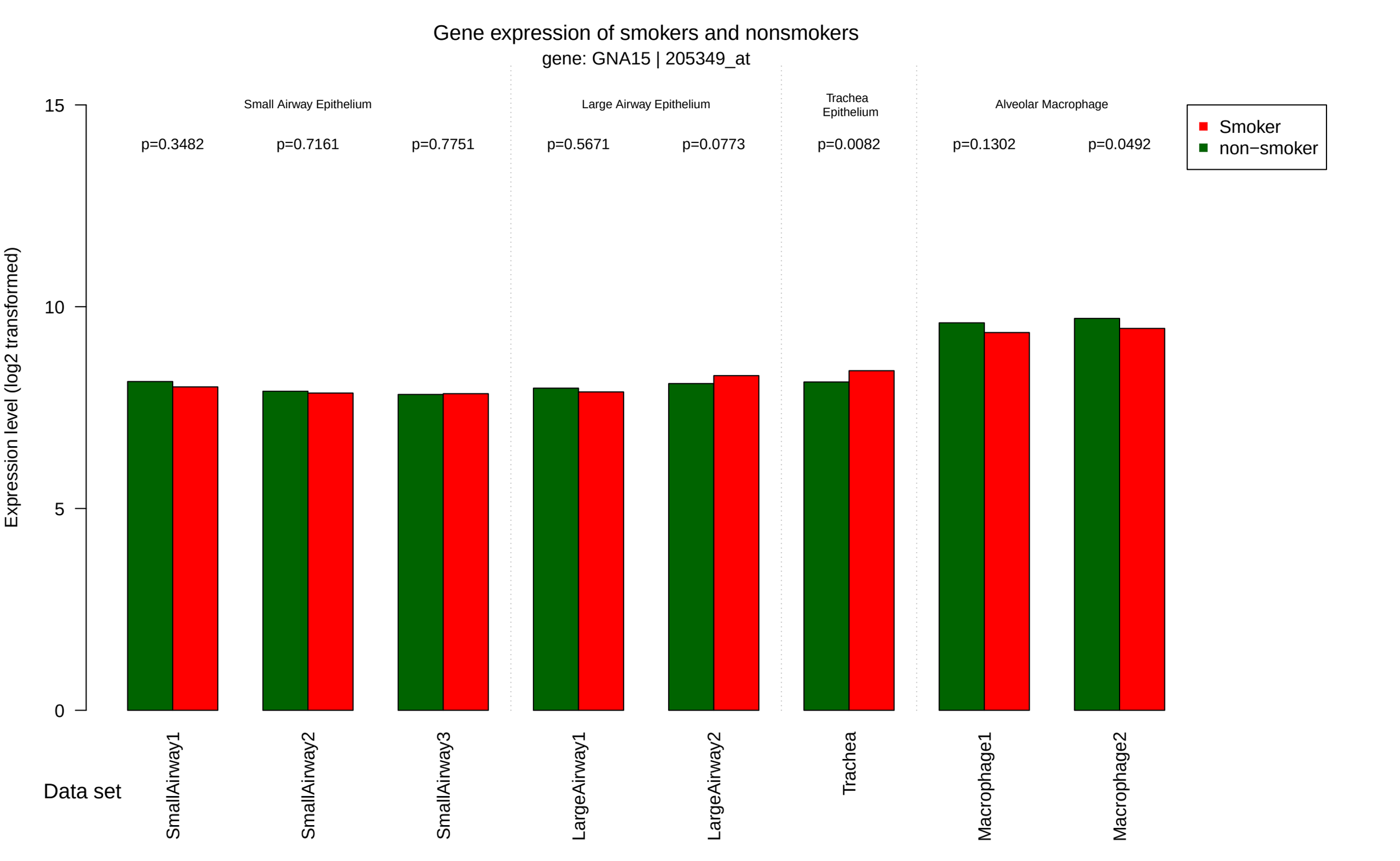
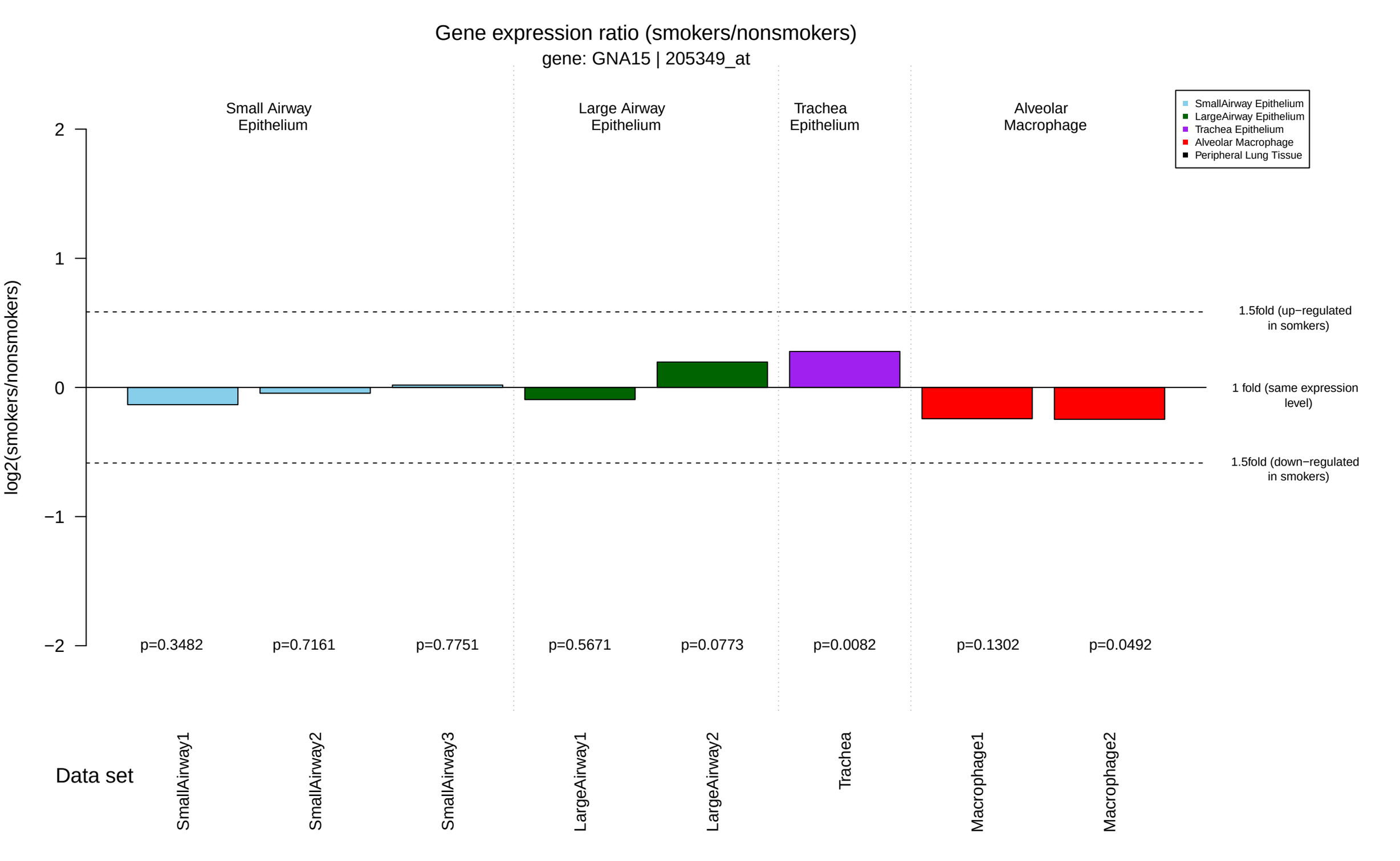
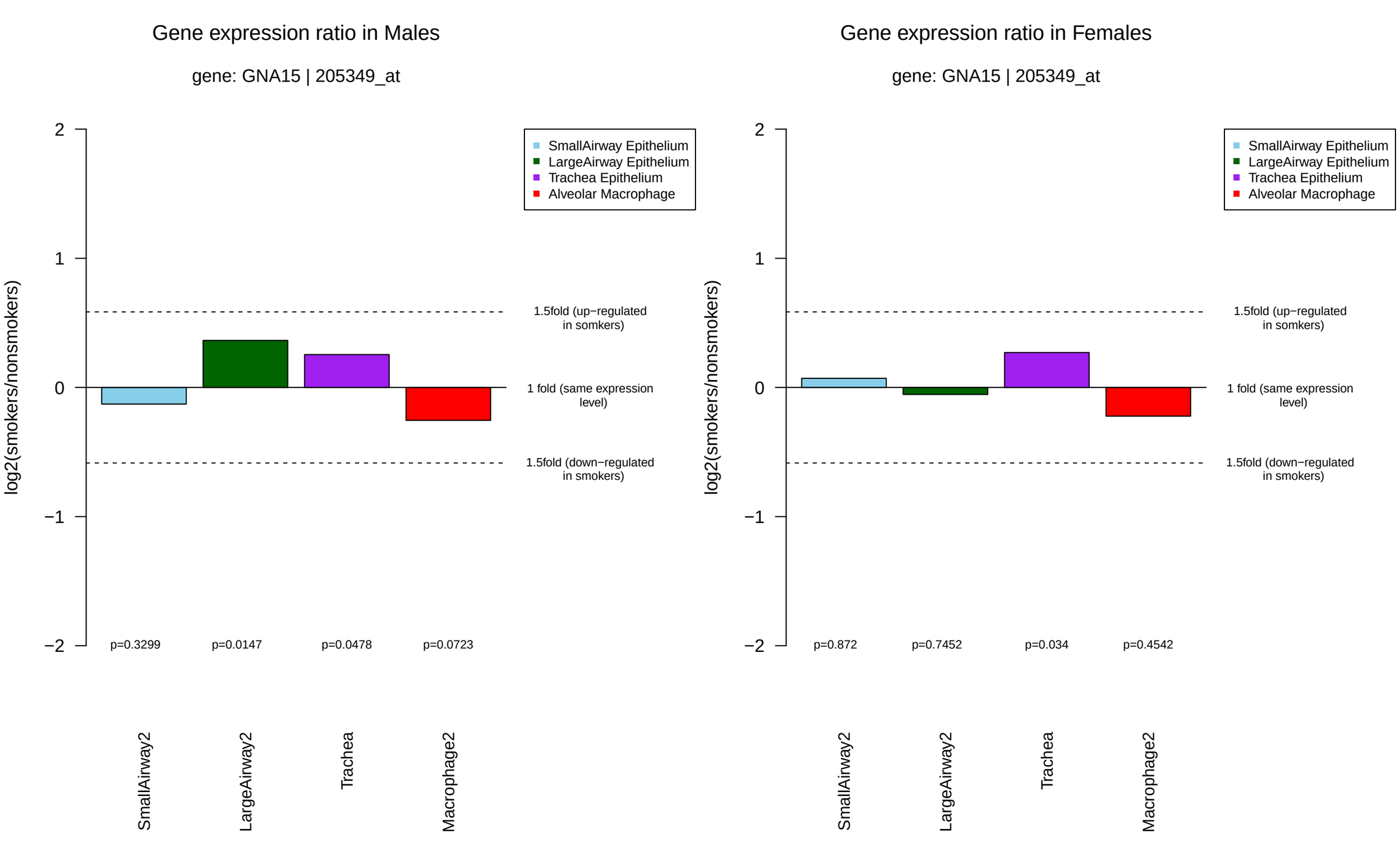
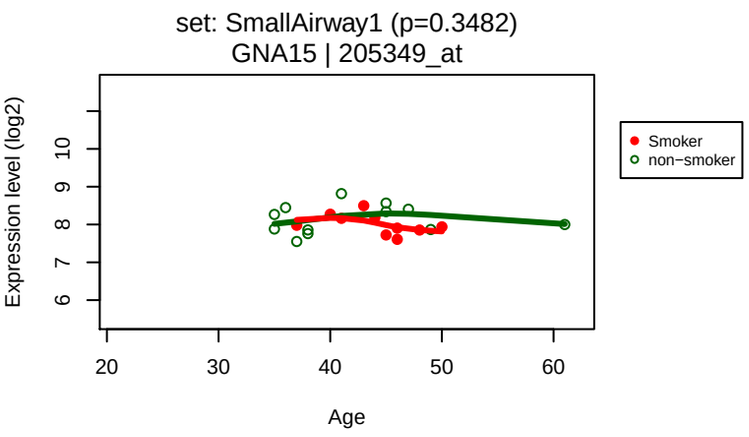
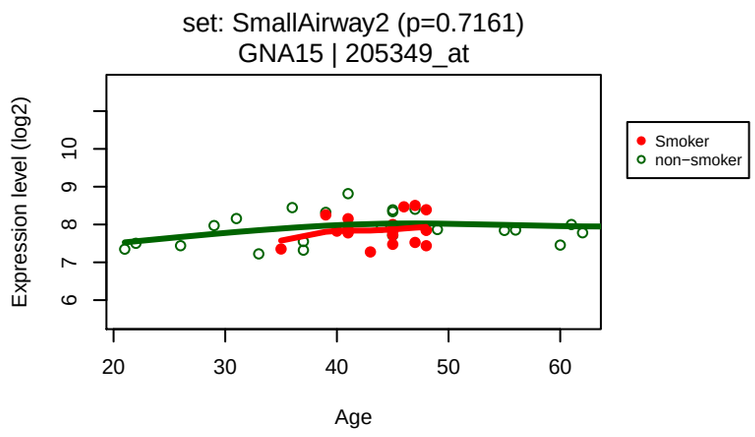
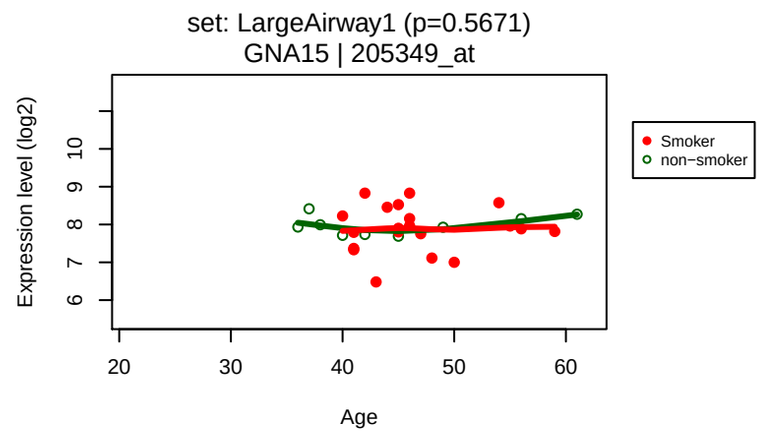

Gene expression of smokers and nonsmokers
gene: GNA15 | 205349_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3482
p=0.7161
p=0.7751
p=0.5671
p=0.0773
p=0.0082
p=0.1302
p=0.0492
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

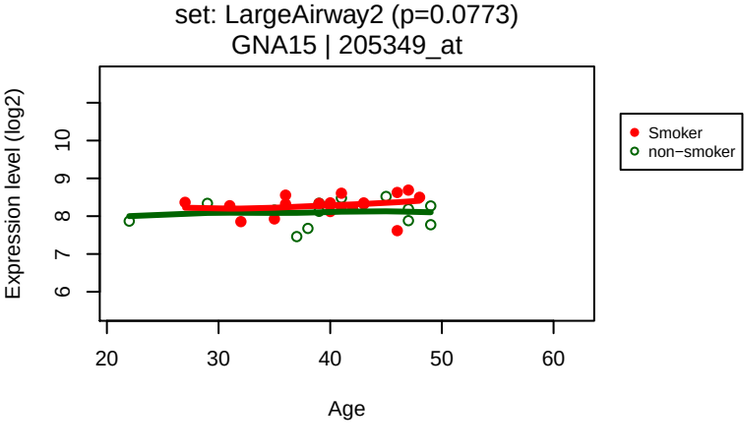

Gene expression ratio (smokers/nonsmokers)
gene: GNA15 | 205349_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3482
p=0.7161
p=0.7751
p=0.5671
p=0.0773
p=0.0082
p=0.1302
p=0.0492
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

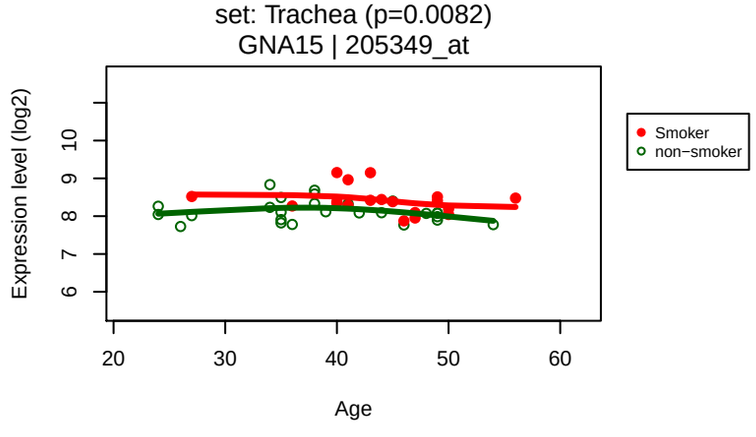

Gene expression ratio in Males
gene: GNA15 | 205349_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3299
p=0.0147
p=0.0478
p=0.0723
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA15 | 205349_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.872
p=0.7452
p=0.034
p=0.4542
SmallAirway2
LargeAirway2
Trachea
Macrophage2

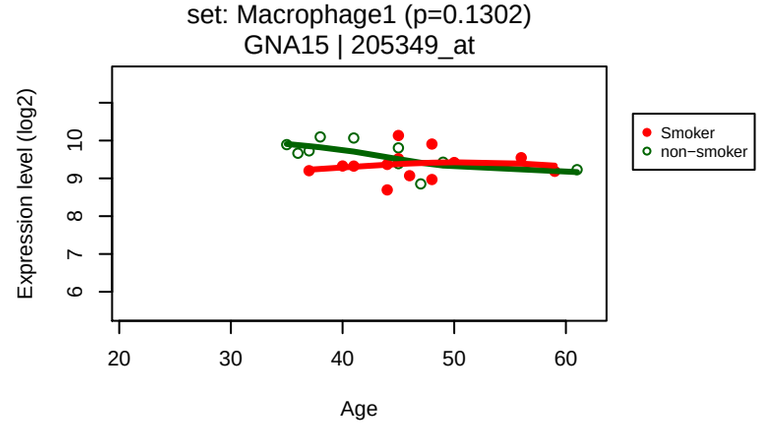

set: SmallAirway1 (p=0.3482)
GNA15 | 205349_at
Expression level (log2)
Age
Smoker
non−smoker

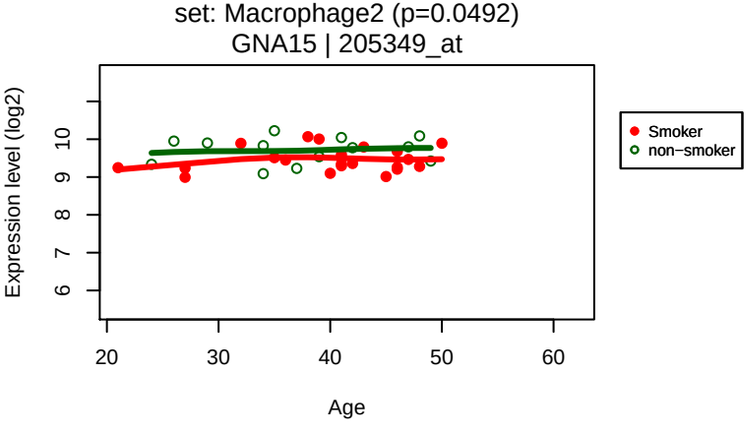

set: SmallAirway2 (p=0.7161)
GNA15 | 205349_at
Expression level (log2)
Age
Smoker
non−smoker

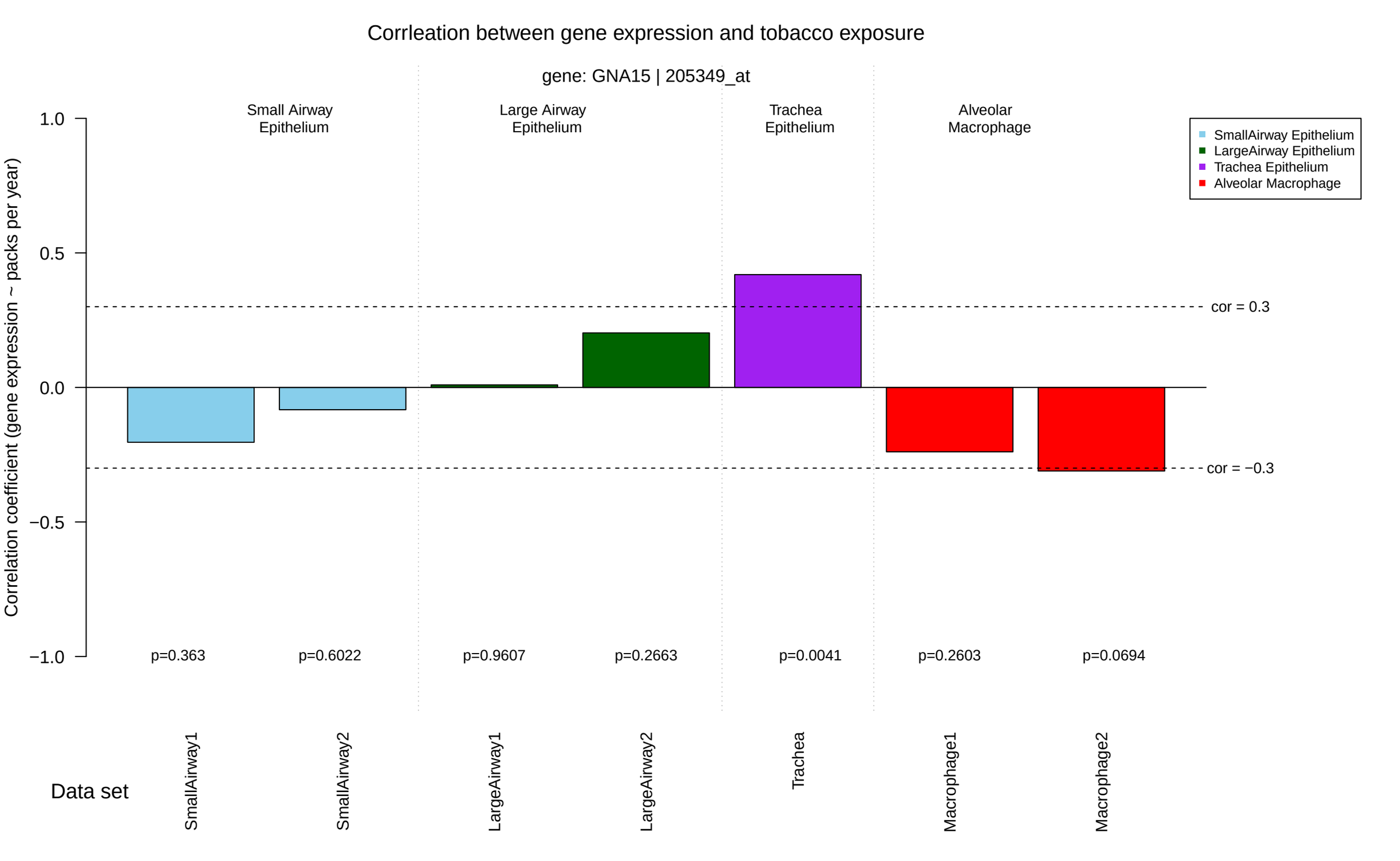

set: LargeAirway1 (p=0.5671)
GNA15 | 205349_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0773)
GNA15 | 205349_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0082)
GNA15 | 205349_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1302)
GNA15 | 205349_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0492)
GNA15 | 205349_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA15 | 205349_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.363
p=0.6022
p=0.9607
p=0.2663
p=0.0041
p=0.2603
p=0.0694
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2